handy for this. Since both of these are really tough on the processor and GPU, we run them in succession and examine how hot the processor is. There are apps like CPU-Z and CPU-Temperature that help us note the temperature before and after the tests.

Next, we play a game (that remains constant for all phones) on the phone, for 15 minutes. You can choose how long you play for, but 15 minutes is generally the time taken for even the best processors to heat up and throttle themselves. The temperature is noted before and after this. We consider 45-50 degrees to be acceptable, while anything above that is worrisome.

While doing all this, you must remember to keep the temperature conditions in your room the same for all smartphones that you are testing. If the air conditioner is at 26 degrees, then it must be so whenever you're testing a smartphone.

Storage: Remember the part of the AnTuTu test we said we would talk about later? This is it. Most smartphone reviews talk about how

much storage and phone has and whether an SD card slot is provided or not, but that should never be the end of it.

Android users especially should care about this, since the OS has dismal SD card support to say the least. The flash storage on your phone is essentially an SD card that the OEM fits into the device while building it. Since you, the user, won't be able to reach it, this is also an area where many smartphone makers can cut costs.

The AnTuTu benchmark provides a number for the I/O performance on your smartphone's storage. This is important to figure out what kind of storage you have. Higher the number, better it is.

The inbuilt flash storage is where all of your apps will reside and it can have major effects on your phones overall performance. In fact, you'll notice sometimes phones with good benchmark scores still tend to stutter in performance. This can be because of poor quality of flash storage inside.

Camera: For the camera, there are four basic light conditions that we test in – sunlight, yellow light, indoor fluorescent lights and low light. The first three are pretty easy to achieve, we just have the same subject set up in the light conditions mentioned above. Low light is a little more difficult, simply because it is harder to achieve the same amount of light for all shots. We do this by ensuring that our subjects are kept at a fixed point, where we can control the lights ourselves. None of the subjects used for our camera tests are ever moved.

We also use a colour chart, which gives us a very good idea of how a smartphone's camera performs with various colours. The colours range from the deep blacks to the bright yellows, reds and more. This colour chart is also fixed at one point, where the light conditions can never be changed.

All our camera tests are done on the auto mode, without using any of the smartphone's added camera tweaks. This is because smartphone cameras are meant to be point and shoot cameras, meant for everyday users. If a camera can do well on the auto mode, then it can do so on the other modes as well. We of course test the other modes out as well, but we do that separately, during regular usage.

If a smartphone is too dependent on its camera tweaks, then that in itself is a weakness, since the user may or may not be able to understand said tweaks.

Battery: As mentioned before, a bigger battery doesn't necessarily ensure better battery life. A 2600 mAh battery won't last as long on a 1080p display as it would on a 720p display. It may also differ based on the display size, processor used and other features of the device. So, it is very important to keep a uniform battery test, which will be the same for all devices, irrespective of their specifications. This ensures that all devices are put through the exact same amount of load when testing the battery.

For our in-house tests, we run a 1080p video (same for all phones) with WiFi and all background activity on and the display on full brightness. We run the video for exactly one hour and note the drop in the battery having started the test at 100%. It is important to start all tests at 100%, as the percentage drops reduce when you go down the ladder.

Since modern SoCs are optimized for video, we also use the Geekbench 3 and PC Mark for Android battery tests in order to get an objective view of the battery life.

Call Quality: It is still a phone after all, so a review can not be complete without ensuring proper call quality. For this we check for crackling or any other noise during calls. This is a two way street, meaning things have to be clear on both sides. We also examine the in-call volume, with and without the loudspeaker.

Real World Usage

While all of the above sounds very good, none of that would make sense without real world usage. Our battery tests would simply be numbers, worth nothing, if we couldn't transform them into real world figures. If a phone has 4GB RAM, then we run multiple games on it one after the other and put it through as much intensive usage as we can to see if it can take it. If the Galaxy S6 Edge scores over 60k on AnTuTu, then we put it through a lot of intensive tasks to see how long it takes for the processor to throttle down enough for that rating to drop.

In addition, we use each smartphone as our own primary device, doing our routine day to day activities to check for glitches and issues. We may use the benchmarks for testing, but they are useless if you don't know what they would translate to in real life, which is what we do through real life usage. We test out each and every benchmark number for what we expect it to translate to and don't sit down to write a review before we're convinced that the numbers are accurate. If they are not, then we try to represent the real world usage using benchmarks. All this is done to ensure that neither human, nor machine is making any errors and the resulting review is as accurate as possible.